

Electronics

Electric current is caused by a drift of electrons through a circuit. An electronic device uses electricity in a more precise way than simple electric appliances, to capture digital photos or play your favorite songs.

While it takes a large electric current to boil water, electronics use carefully controlled electric currents thousands or millions of times smaller, and sometimes just single electrons, to operate a range of complex devices. Computers, smartphones, amplifiers, and TV remote controls all use electronics to process information, communicate, boost sound, or switch things on and off.



Smartphone

Cell phones are now so advanced that they are really hand-sized computers. As well as linking to other digital devices, they contain powerful processor chips and plenty of memory to store applications.

Printed circuit board (PCB)

The “brain” of a smartphone is on its printed circuit board—a premanufactured electronic circuit unique to a particular device. The PCB is made from interconnected microchips, each of which is constructed from a tiny wafer of silicon and has an integrated circuit inside it containing millions of microscopic components.

Motherboard

The main printed circuit board, which is the phone’s main processor, is also referred to as a mainboard or logic board.

Electronic components

Electronic circuits are made of building blocks called components. A transistor radio may have a few dozen, while a processor and memory chip in a computer could have billions. Four components are particularly important and appear in nearly every circuit.



Diode

Diodes make electric current flow in just one direction, often converting alternating to direct current.



Resistor

Resistors reduce electric current so it is less powerful. Some are fixed and others are variable.



Transistor

Transistors switch current on and off or convert small currents into bigger ones.



Capacitor

Capacitors store electricity. They are used to detect key presses on touch screens.